

A Hybrid Transformer Architecture for Code-Switching Detection in Filipino-English Text

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Abstract—In multilingual societies, Natural Language Processing (NLP) systems often falter when confronted with the fluid alternation of languages known as code-switching (CSW). In the Philippines, Taglish—a dynamic blend of Tagalog and English—is a pervasive form of communication across formal and informal domains, shaped by bilingual education and media exposure. Despite its ubiquity, Taglish remains underrepresented in NLP research, and existing models trained on monolingual data struggle with its syntactic irregularity and sociolinguistic nuance. Grounded in linguistic theory and sociocultural context, this study explores a hybrid transformer architecture that integrates XLM-RoBERTa’s multilingual contextual embeddings with BART’s sequence reconstruction capabilities, further enriched with part-of-speech (POS) information. A domain-balanced corpus of 5,201 Taglish sentences—annotated for token-level language and POS tags—was constructed from social media, conversational exchanges, and spoken transcripts. Experimental results show that the hybrid model improves token-level code-switching detection across varied domains, outperforming baseline and standalone transformer models. Error analysis reveals persistent challenges in handling informal discourse markers and domain-specific noise. The findings contribute to ongoing efforts to build inclusive and linguistically aware NLP systems for low-resource, code-switched languages.

Index Terms—Code-switching, Taglish, multilingual NLP, hybrid transformers, low-resource languages

I. INTRODUCTION

In the field of Natural Language Processing (NLP), understanding multilingual communication presents intricate challenges, particularly in contexts where speakers frequently switch between languages within a single sentence or phrase. This linguistic behavior, known as code-switching (CSW), disrupts assumptions of language uniformity and contextual coherence upon which most state-of-the-art NLP models are built. Typically trained on monolingual datasets, these models struggle when confronted with the complex, context-driven alternation of languages found in naturally occurring discourse. As Poplack [1] emphasized, code-switching is not random but governed by syntactic and sociolinguistic patterns that NLP systems still struggle to replicate.

One striking example of this phenomenon is Taglish—a fluid blend of Tagalog and English—widely spoken across the Philippines. Taglish serves not only as a linguistic hybrid but also as a cultural marker, reflecting the Philippines’ complex colonial history and bilingual educational system [2],

[3]. Young and Igcailinos [4] note that bilingual proficiency is deeply embedded in Filipino education, while a recent survey by Social Weather Stations [5] found that nearly half of Filipinos are comfortable thinking in English, and the majority are fluent in both languages. This dual proficiency underpins Taglish’s widespread use across domains: in science classrooms [6], over 58% of utterances exhibit intrasentential switching [7], and in online learning environments [8], students commonly blend languages for greater expressive flexibility.

Yet this very fluidity poses significant challenges for NLP systems. Models trained on high-resource monolingual corpora frequently misinterpret transitions between languages, resulting in decreased accuracy for language identification, part-of-speech tagging, and downstream tasks [9], [10]. The scarcity of annotated datasets for Filipino-English code-switching further exacerbates this issue [11], [12]. While transformer-based models such as XLM-RoBERTa [13] and BART [14] offer promising avenues for multilingual NLP, they require domain-specific adaptation to capture the nuanced structure of Taglish. As Huzaifah et al. [15] and Zhang et al. [16] observe, even large multilingual models continue to struggle with fine-grained syntactic and semantic variations in code-switched text.

Building upon linguistic insights [17], [18] and recent advances in transformer architectures, this study proposes a hybrid model that integrates XLM-RoBERTa’s contextual embeddings with BART’s sequence reconstruction capabilities, enriched with part-of-speech (POS) tagging. To support this, a domain-diverse annotated corpus of 5,201 Taglish sentences was constructed, spanning social media posts, conversational exchanges, and spoken transcripts. The models were evaluated on a domain-sensitive test set, with results analyzed at both the token level and across linguistic domains.

The contributions of this work are as follows:

- The creation of a balanced, token-annotated Taglish corpus with language and POS tags, reflecting real-world usage across multiple domains.
- The design of a hybrid transformer architecture that fuses XLM-RoBERTa and BART to enhance code-switching detection.

- Empirical evaluation of the model’s performance across social media, conversational, and spoken domains, with in-depth error analysis.
- Broader insights into the applicability of hybrid architectures for low-resource, code-switched languages beyond Taglish.

The remainder of this paper is organized as follows: Section II reviews related work on code-switching detection and transformer-based models. Section III details the dataset construction, model architecture, and experimental setup. Section IV presents the results and discusses key findings. Section V concludes the study and outlines directions for future research.

II. RELATED WORK

A. Theoretical Frameworks on Code-Switching

Code-switching, the alternating use of two or more languages within a sentence or conversation, has long been the subject of linguistic inquiry. Early foundational models such as Poplack’s Constraints Model [1] demonstrated that code-switching is governed by syntactic regularities rather than occurring at random. Poplack identified consistent patterns in the placement of switch points, with syntactic boundaries playing a critical role.

Building on this, Myers-Scotton’s Matrix Language Frame (MLF) model [17] introduced the concept of a dominant *matrix* language that provides the grammatical structure, while an *embedded* language contributes lexical material. In the context of Taglish, Tagalog often functions as the matrix language, with English insertions varying from isolated words to entire phrases.

Muysken’s typology [18] further categorized code-switching into insertion, alternation, and congruent lexicalization. Taglish exhibits all three types, with insertion and congruent lexicalization particularly prevalent in informal domains such as social media [19]. These linguistic models not only inform sociolinguistic understanding but also provide valuable perspectives for computational modeling, especially when constructing annotation guidelines and designing model architectures.

B. Challenges in NLP for Code-Switching

While the theoretical underpinnings of code-switching are well established, the practical task of processing code-switched text in NLP remains a significant challenge. NLP systems trained on monolingual corpora typically assume language consistency and struggle with the syntactic fluidity and irregular alternation patterns inherent in code-switched text [9], [10].

A key obstacle is the scarcity of high-quality, token-level annotated datasets, particularly for low-resource language pairs such as Filipino-English [11], [12]. Without sufficient training data, models exhibit reduced performance in fundamental tasks such as language identification and part-of-speech tagging. Moreover, Out-of-Vocabulary (OOV) words—including localized slang and hybrid expressions—further complicate modeling efforts [12].

Recent studies have shown that even large multilingual transformer models face limitations in handling code-switched

inputs. Huzaifah et al. [15] and Zhang et al. [16] found that models such as XLM-R and GPT-based architectures struggle with fine-grained syntactic and semantic variation, particularly when switching occurs within phrases or clauses.

In speech processing, similar challenges have been observed. Mustafa et al. [20] reported notable drops in automatic speech recognition accuracy in the presence of mid-utterance code-switching. These findings underscore the need for domain-adapted models and data-driven approaches that explicitly account for the unique characteristics of code-switched language.

C. Evolution of Code-Switching Detection Approaches

Early computational approaches to code-switching detection relied on rule-based systems and statistical models. Unigram and bigram frequency counts [21], part-of-speech-based pattern matching [22], and dictionary-based filters were initially employed to identify switch points. While effective in constrained domains, these approaches struggled with informal, noisy, and domain-diverse text.

Subsequent research explored machine learning techniques such as Conditional Random Fields (CRFs), Support Vector Machines (SVMs), and neural networks. Choudhury et al. [23] introduced curriculum learning to mitigate performance degradation when transferring from monolingual to code-switched data. Bilingual embeddings [24] and adversarial training [25] further improved model robustness.

The advent of transformer architectures marked a significant advancement. Models such as BERT, XLM-RoBERTa [13], and BART [14] introduced self-attention mechanisms capable of capturing long-range dependencies and cross-lingual contextual information. XLM-R, trained on multilingual corpora using masked language modeling and translation language modeling objectives, has proven effective in multilingual classification tasks. BART’s sequence reconstruction capabilities enable improved handling of noisy and irregular input—features highly relevant to code-switched text.

However, even these advanced models exhibit limitations when applied to code-switching, particularly in low-resource and domain-variable contexts. Winata et al. [26] and Zhang et al. [16] emphasize the need for targeted fine-tuning and hybrid architectures that can leverage both contextual embeddings and sequence modeling capabilities.

D. Progression of Taglish Code-Switching Detection

Within the Filipino NLP community, research on Taglish code-switching detection has evolved from early pattern-based methods to transformer-based approaches. Bacatan et al. [21] demonstrated that basic n-gram models could identify switch points in small datasets but faced scalability issues. Oco and Roxas [22] integrated part-of-speech tagging to enhance accuracy, though intra-word and informal switching remained challenging.

The introduction of the TweetTaglish dataset [27] provided a valuable resource for benchmarking Taglish models. Herrera

et al. developed baseline classifiers using this corpus, highlighting both the potential and limitations of existing models in handling informal social media text.

Recent work by Cosme and De Leon [28] applied XLM-RoBERTa to sentiment analysis on code-switched Filipino-English reviews, achieving improved performance over lexicon-based methods. However, their study also noted the persistent difficulty of handling domain-specific and informal language variation.

Despite these advancements, key gaps remain. Most prior studies focus on narrow domains, lack domain-sensitive evaluation, and do not explore hybrid architectures that combine contextual and generative modeling. The present study addresses these gaps by proposing a hybrid XLM-R + BART model, trained on a domain-diverse, token-annotated Taglish corpus, with the goal of improving both detection accuracy and robustness across linguistic domains.

III. METHODOLOGY

A. Dataset Construction and Annotation

To support the development of robust models for Taglish code-switching detection, a domain-diverse annotated corpus was constructed. The final dataset comprises 5,201 sentences and 54,956 tokens, covering a wide range of syntactic and contextual variation in naturally occurring Taglish.

The corpus was sourced from three primary domains:

- **Social Media:** Informal content from the TweetTaglish dataset [27], rich in lexical mixing, abbreviations, and slang.
- **Spoken Transcripts:** Semi-formal utterances transcribed from publicly available YouTube podcasts, capturing fluent intrasentential switching in structured discourse.
- **Conversational Exchanges:** Simulated chat-based dialogues drawn from the LazadaQA-Taglish dataset [29], reflecting pragmatic switching in customer interactions.

Each token in the dataset was annotated with two types of linguistic information:

- **Language Tagging:** Tokens were labeled as TAG (Tagalog), ENG (English), or OTH (Other), where OTH includes emojis, symbols, and named entities.
- **Part-of-Speech (POS) Tagging:** POS tags were assigned based on a simplified version of the Universal POS Tagset.

Annotation was conducted by three bilingual annotators, with inter-annotator agreement achieving a Cohen's Kappa score of 0.83 [30], indicating substantial agreement. Annotation discrepancies were resolved through consensus sessions with the support of a supervising linguist.

B. Model Architectures

The study evaluates three model architectures of increasing complexity:

- **Baseline Model:** Character n-gram features combined with logistic regression, serving as a lightweight benchmark.

- **XLM-RoBERTa:** A transformer-based token classifier leveraging XLM-RoBERTa [13], fine-tuned for multilingual contextual representation.
- **Hybrid XLM-R + BART:** A novel architecture integrating XLM-RoBERTa embeddings with BART [14] sequence representations, further enriched with POS embeddings.

C. Hybrid Fusion Mechanism

The hybrid model was designed to capitalize on the complementary strengths of XLM-RoBERTa and BART. While XLM-RoBERTa provides robust multilingual contextual embeddings, BART offers strong capabilities for sequence reconstruction and fluency modeling — particularly valuable in handling the syntactic irregularity of code-switched text.

As illustrated in Figure 1, the fusion mechanism operates by concatenating the token-level embeddings produced by the XLM-R encoder and the BART encoder. A joint attention mechanism is applied to the concatenated representations, allowing the model to dynamically weigh the contributions of each encoder for downstream classification.

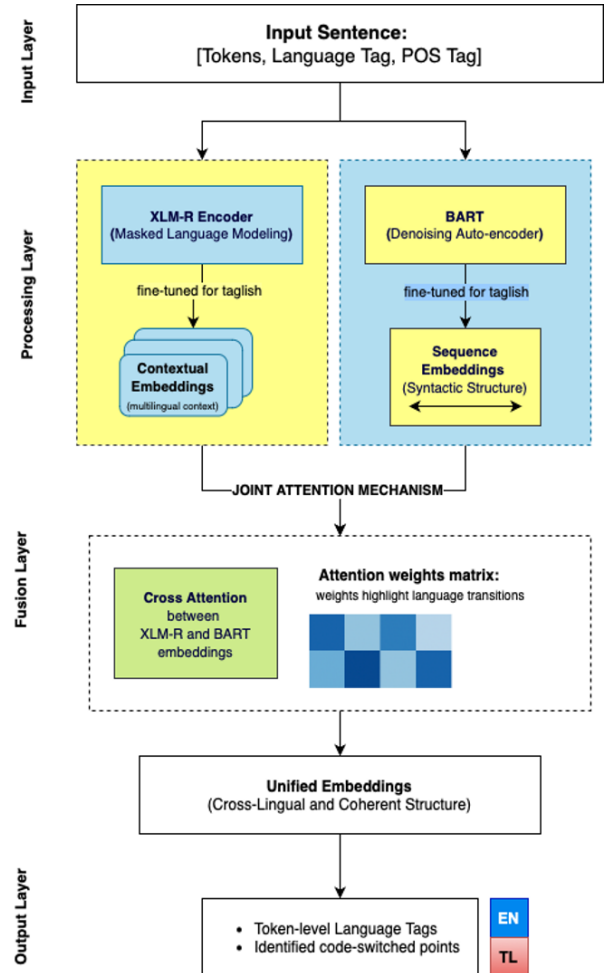


Fig. 1. Fusion layer combining XLM-RoBERTa embeddings with BART sequence embeddings via joint attention.

The resulting hybrid representations are passed through a classification head to predict token-level language labels. POS tag embeddings are incorporated by concatenating them with the encoder outputs prior to joint attention, providing additional syntactic cues.

D. Experimental Setup and Evaluation Protocol

A rigorous experimental framework was adopted to assess model performance across domains. The dataset was partitioned as follows:

- **Training Set:** 80% of annotated data.
- **Validation Set:** 10% of annotated data.
- **Domain-Specific Test Set:** 10% reserved for domain-sensitive evaluation, comprising 1,000 sentences each from social media, spoken transcripts, and conversational exchanges.

The models were evaluated using standard NLP metrics: token-level precision, recall, F1-score, and accuracy. Additionally, code-switching specific analyses were performed to assess the models' ability to detect switch points and handle challenging cases such as informal discourse markers and noisy tokens.

As illustrated in Figure 2, the experimental pipeline encompasses data preprocessing, model training and evaluation, and prototype deployment. The hybrid model variants were fine-tuned using learning rates and dropout rates optimized through grid search, with performance trends monitored across training epochs.

This comprehensive methodology enables a thorough investigation of the strengths and limitations of hybrid transformer architectures in the context of Taglish code-switching detection.

IV. RESULTS AND DISCUSSION

A. Experimental Framework

The experimental framework evaluated five model configurations for token-level Taglish code-switching detection:

- Baseline: Character n-gram + logistic regression.
- XLM-RoBERTa (XLM-R).
- XLM-R + POS.
- Hybrid XLM-R + BART.
- Hybrid XLM-R + BART + POS.

All models were trained and evaluated on the domain-diverse annotated corpus described in Section III. The domain-sensitive test set included 1,000 sentences each from social media, spoken transcripts, and conversational exchanges. Evaluation metrics included token-level precision, recall, F1-score, and accuracy.

B. Model Evaluation and Comparison

Table I summarizes the overall performance of each model configuration.

The Hybrid XLM-R + BART model achieved the highest overall performance, with an F1-score of 0.94 and accuracy of 0.95. The addition of POS features provided modest gains in certain cases but did not consistently outperform the hybrid model without POS.

TABLE I
SUMMARY OF MODEL PERFORMANCE (TOKEN-LEVEL METRICS).

Model	Accuracy	Precision	Recall	F1-Score
Baseline	0.51	0.98	0.51	0.66
XLM-R	0.84	0.83	0.84	0.83
XLM-R + POS	0.79	0.80	0.79	0.77
Hybrid	0.95	0.93	0.95	0.94
Hybrid + POS	0.92	0.90	0.92	0.91

C. Per-Class Behavior and Token-Level Analysis

Figure 3 illustrates the per-class F1-scores for each model across the three language tags: TAG, ENG, and OTH.

The Hybrid model achieved high F1-scores for both TAG and ENG classes, with a slight drop observed in the Hybrid + POS variant for ENG. The OTH class remained the most challenging across all models, reflecting the heterogeneity of this category (emojis, named entities, symbols).

D. Domain Sensitivity Evaluation

The domain-sensitive evaluation revealed variations in model performance across social media, spoken transcripts, and conversational text. Table II summarizes the domain-level F1-scores for each model.

TABLE II
F1-SCORES OF EACH MODEL ACROSS DOMAINS.

Model	Social Media	Spoken	Conversational
Baseline	0.66	0.64	0.68
XLM-R	0.78	0.79	0.81
XLM-R + POS	0.76	0.78	0.82
Hybrid	0.82	0.83	0.85
Hybrid + POS	0.81	0.84	0.86

Conversational text yielded the highest F1-scores across all models, likely due to clearer sentence boundaries and reduced noise. Social media remained the most challenging domain, reflecting the complexity of user-generated content.

E. Error Analysis and Token Confusions

A detailed error analysis highlighted several persistent challenges:

- Short discourse markers such as *na*, *naman*, and *eh* were frequently misclassified, particularly when adjacent to English tokens.
- The OTH class showed low recall across all models, as it included diverse token types with irregular patterns.
- Informal discourse markers and stylized abbreviations in social media contributed to higher confusion rates in this domain.

Figure 4 presents a sample confusion matrix for the best-performing Hybrid + POS model.

F. Hyperparameter Tuning and Performance Trends

Hyperparameter tuning revealed several trends:

- The Hybrid model benefited from longer training durations, with performance gains observed up to 10 epochs.

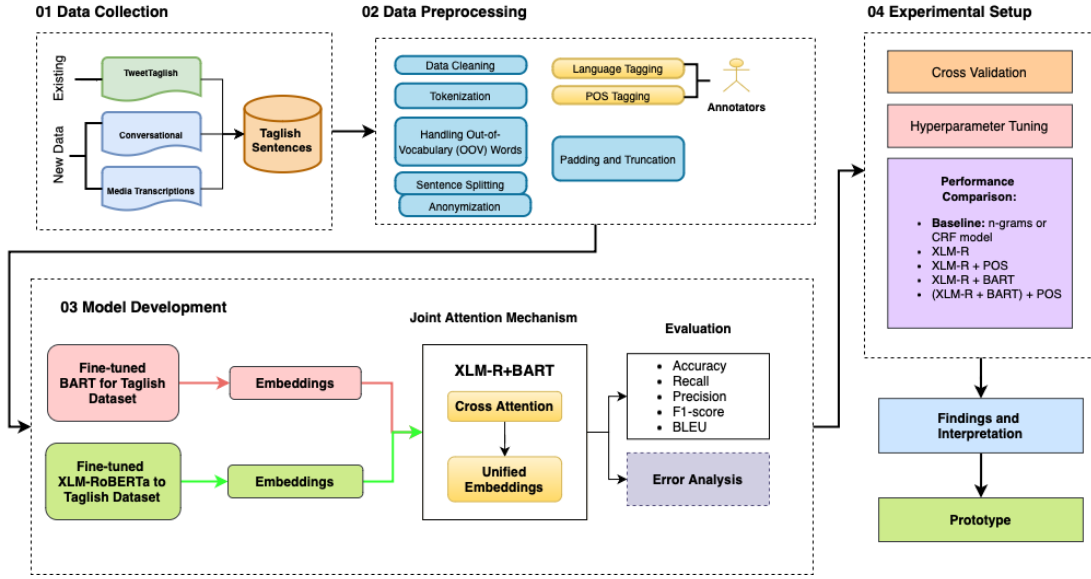


Fig. 2. Framework for Taglish code-switching detection and analysis.

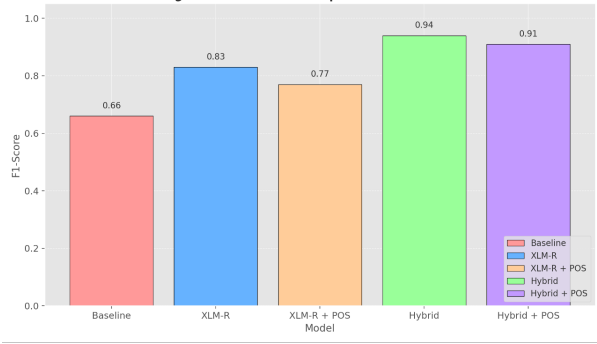


Fig. 3. Per-class F1-scores across model configurations.

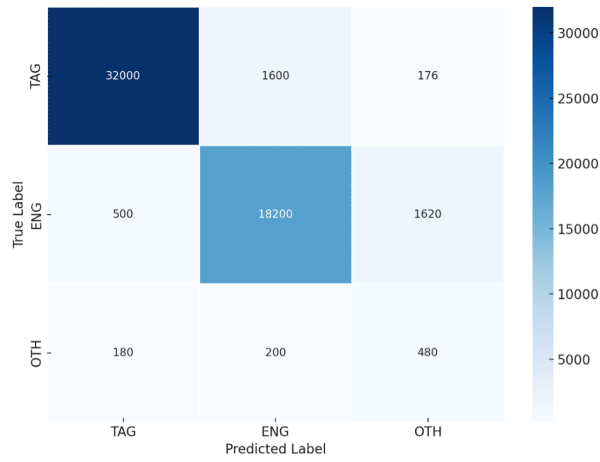


Fig. 4. Confusion matrix for Hybrid + POS model.

- POS integration improved performance in Hybrid models, particularly in conversational and spoken domains.
- Lower dropout rates (0.1) outperformed higher values across XLM-R configurations.

Figure 5 illustrates the accuracy trends across training epochs for key model configurations.

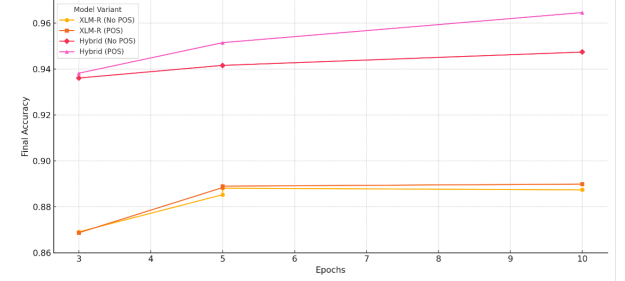


Fig. 5. Training accuracy trends across epochs.

G. Generalizability Insights

The study's findings suggest that hybrid transformer architectures are effective in handling the syntactic and semantic complexity of Taglish code-switched text. The domain-based evaluation highlights the importance of training on linguistically diverse data to enhance model robustness. Moreover, the hybrid approach demonstrated potential applicability to other low-resource, code-switched languages, provided that appropriate domain adaptation and fine-tuning are performed.

V. CONCLUSION AND FUTURE WORK

This study explored hybrid transformer architectures for token-level code-switching detection in Taglish, a linguistically dynamic and low-resource mixed language. By integrating XLM-RoBERTa embeddings with BART's sequence

representations, the proposed model achieved improved detection performance across social media, spoken transcripts, and conversational text. The domain-diverse Taglish corpus developed for this work enabled a comprehensive evaluation of model generalizability, while error analysis highlighted ongoing challenges in processing informal discourse markers and heterogeneous OTH tokens.

Future directions include expanding the annotated corpus to cover additional domains and linguistic variations, refining the hybrid architecture to better handle informal content, and exploring cross-linguistic transfer to other Philippine language pairs. Further work will also address the computational efficiency of the hybrid model, facilitating its deployment in practical low-resource settings. These efforts contribute to the broader goal of advancing inclusive NLP systems capable of handling the complexity of multilingual and code-switched communication. Further work will also explore practical deployment of the hybrid model through user-facing prototypes for educational and research applications.

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